

LSci 51/Psych 56L: Acquisition of Language

Lecture 19

Language in special populations I

Announcements

Review questions available for language development in special populations

HW6 due 9/8/25

Please fill out course evaluations

Consider taking more language science classes (LSci)!



Special populations



Why special populations?

Not everyone is a typically-developing child.

We can explore how different human abilities contribute to the human language acquisition process.

Does language develop differently if there's no visual input (blind children**)?**



Does language develop differently if there's no auditory input (deaf children**)?**



Blind children



Why blind children?

Blind children hear and talk, but lack visual cues to language:

Ex 1: achieving joint attention through pointing and eye gaze isn't possible.



Why blind children?

Blind children hear and talk, but lack visual cues to language:

Ex 2: visual information about lip configurations for producing sounds isn't available.



Linguistic development of blind children

Phonological development: Blind children make **more errors** than sighted children with sounds that involve visible articulatory movements (/b/, /m/, /f/).



Linguistic development of blind children

Lexicon development: Blind children have **fewer words for things that can be seen, but not touched** (like *flag*, *moon*). They have **more words for things associated with auditory change**.



Notably, Bedny, Koster-Hale, Elli, Yazzolino, & Saxe (2019) show that blind individuals have **detailed knowledge of visual perception verbs** like *peek*, *stare*, *blaze*, *glow*, and *flash* — **just as detailed as sighted individuals do**.

Linguistic development of blind children

Syntactic development: Same as that of sighted children.

- Some differences due to mother's input (fewer questions, more commands), which leads to late auxiliary verb (*has*, *is*) acquisition



Insight into first language acquisition

One perspective: language development builds on nonverbal communication, and on accessing the meanings of sentences from the observable nonlinguistic context.

But blind children can't do either of these - yet they still acquire language the same way (and at the same time) as sighted children do.



Implication: Nonlinguistic cues are helpful, but not necessary. Syntactic information in the language itself can be just as useful. (Remember how useful syntactic bootstrapping was for lexical acquisition.)

Deaf children

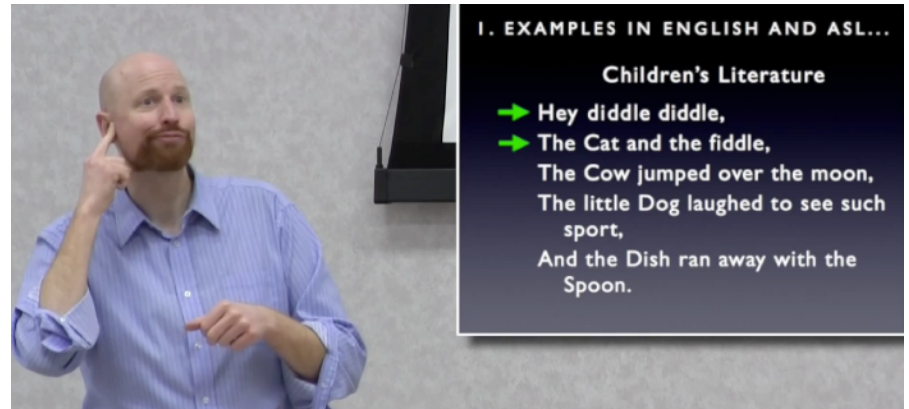


Signed language

Signed languages, which are a main way that deaf individuals communicate, are just as complex as spoken languages - it's just that they're expressed with manual gestures and facial expressions, rather than spoken words.

Sign language sample:

<https://www.youtube.com/watch?v=rloFpxAo93U>



I. EXAMPLES IN ENGLISH AND ASL...

Children's Literature

- Hey diddle diddle,
- The Cat and the fiddle,
- The Cow jumped over the moon,
- The little Dog laughed to see such sport,
- And the Dish ran away with the Spoon.

Signed language is similar to spoken language

Berent, Bat-El, Brentari, Dupuis, & Vaknin-Nusbaum 2017, Brookshire, Lu, Nusbaum, Goldin-Meadow, & Casasanto 2017, Blanco-Elorrieta, Kastner, Emmorey, & Pylkkänen 2018, Trettenbrein, Papitto, Friederici, & Zaccarella 2020: Lots of **similarities in how our brains process** signed and spoken language input

<https://www.sciencedaily.com/releases/2016/11/161108100759.htm>

<https://www.sciencedaily.com/releases/2017/06/170608145521.htm>

<https://www.sciencedaily.com/releases/2018/04/180403085042.htm>

<https://www.sciencedaily.com/releases/2021/02/210219124236.htm>



Signed language

[Extra]

Signed vs. spoken languages:

http://www.youtube.com/watch?v=p_AAttEQj88

(~6 minutes)

(0:37 - 4:12)

Using signed language to identify what the core properties of any language system are

(4:12 - end) [Extra]

Language processing in brains of deaf people (left hemisphere specialization)

Signed language

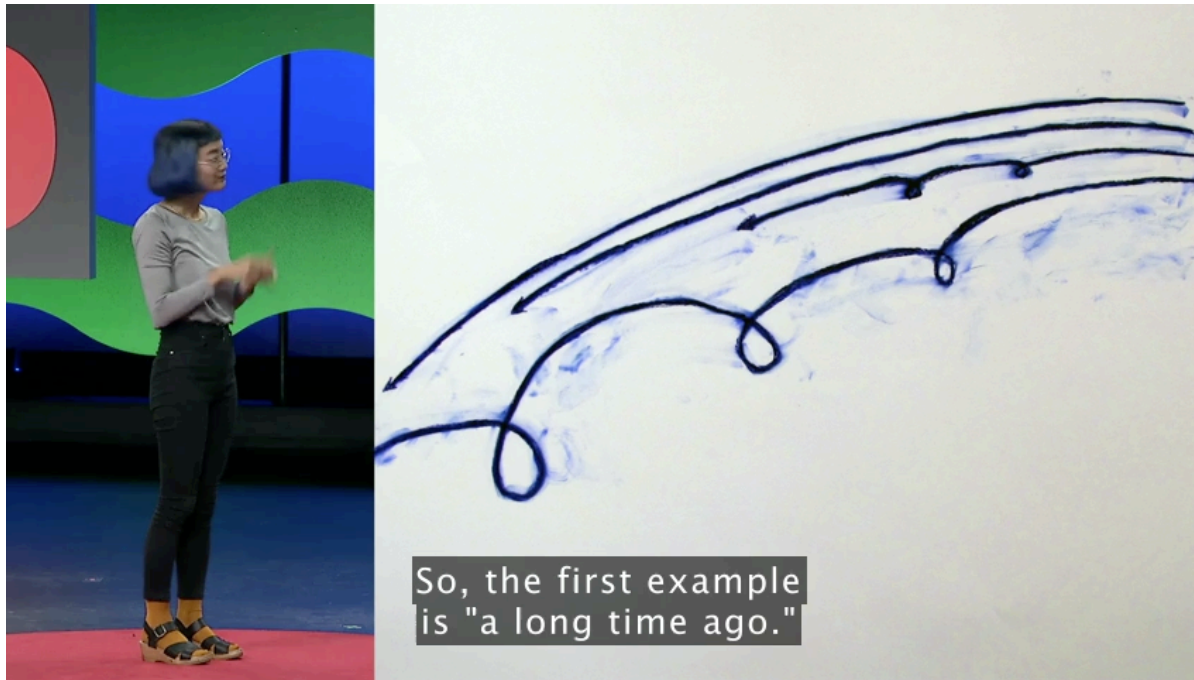
Lillo-Martin & Gajewski 2014:

“In large part, the linguistic analysis of sign languages has led to the conclusion that **universal characteristics of language** can be stated at an abstract enough level to **include languages in both spoken and signed modalities**. For example, languages in both modalities display hierarchical structure at sub-lexical and phrasal level, and recursive rule application. However, this **does not mean that modality-based differences between signed and spoken languages are trivial.**”



Signed language

Signed languages like ASL do have some **iconicity**, where the signs resemble the concepts they represent. This can be particularly useful for conveying additional information about those concepts (Schlenker 2018, <https://www.sciencedaily.com/releases/2018/11/181106104221.htm>).



https://www.ted.com/talks/christine_sun_kim_the_enchanting_music_of_sign_language

11:12-11:51

Sign language features

Like spoken language sounds, sign language signs can be broken into features which can be combined:

- handshape
- palm orientation (direction palm is facing)
- location
- motion



Notably, features in spoken languages combine to form individual sounds (ex: +stop, +voice, +velar = /g/). Features in signed languages combine to form the equivalent of words.

Sign language features make up words [Extra]

<https://www.youtube.com/watch?v=rloFpxAo93U>

0:14 - 3:55

Rhyming in ASL requires breaking the words into their features and playing with those features the way hearing languages break up the sounds of words and play with the sounds



I. EXAMPLES IN ENGLISH AND ASL...

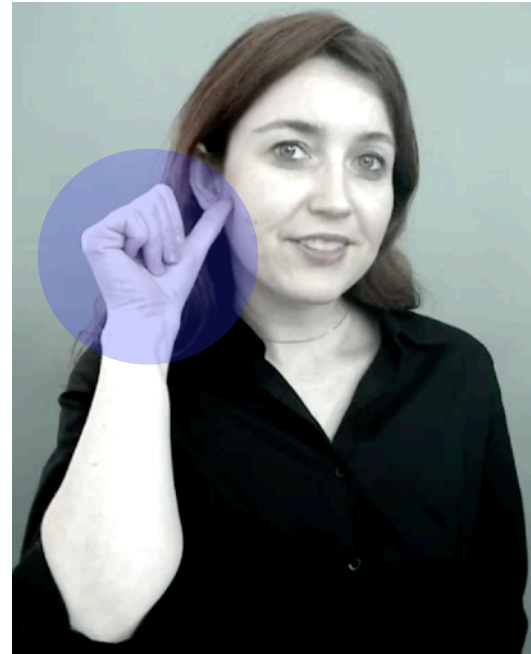
Children's Literature

- Hey diddle diddle,
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The Cow jumped over the moon,
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ASL signs differing only by handshape

<https://www.handspeak.com/word/search/index.php?id=1046>

<https://www.handspeak.com/word/search/index.php?id=2444>

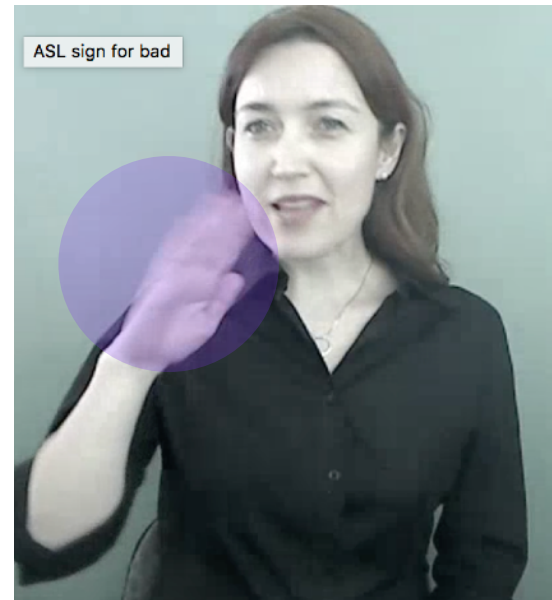
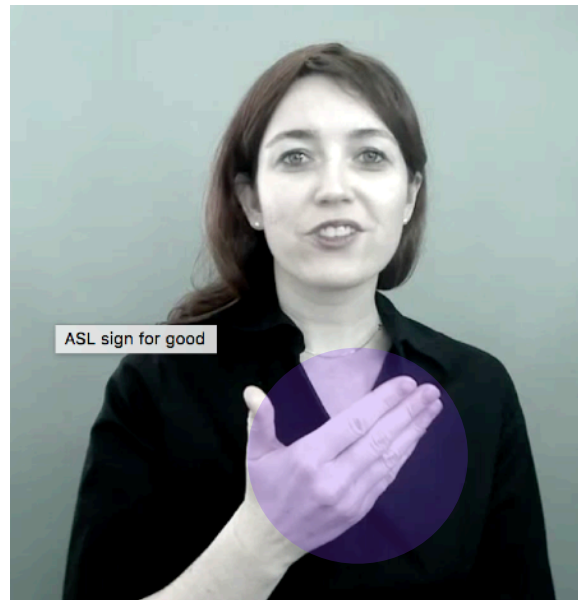


Ex: HOME vs. YESTERDAY

ASL signs differing only by palm orientation

<https://www.handspeak.com/word/search/index.php?id=926>

<https://www.handspeak.com/word/search/index.php?id=150>

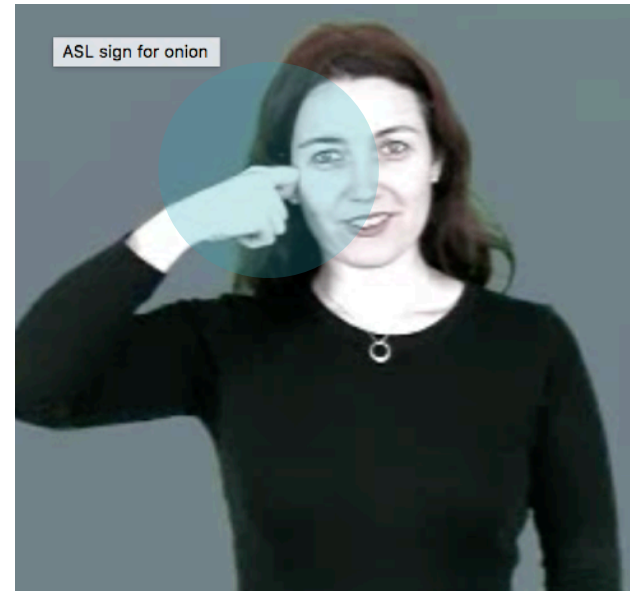
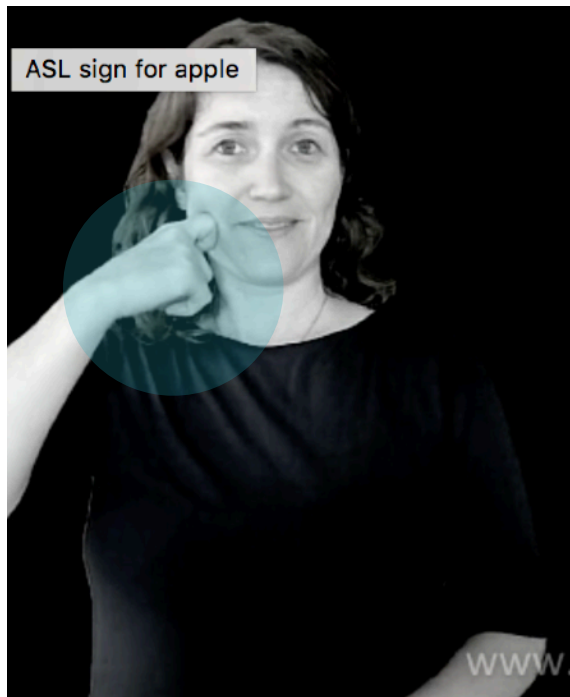


Ex: GOOD vs. BAD

ASL signs differing only by location

<https://www.handspeak.com/word/search/index.php?id=97>

<https://www.handspeak.com/word/search/index.php?id=1559>

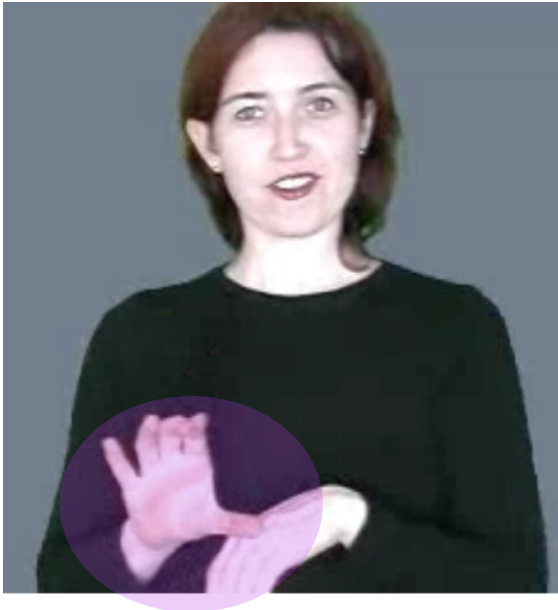


Ex: APPLE vs. ONION

ASL signs differing only by motion

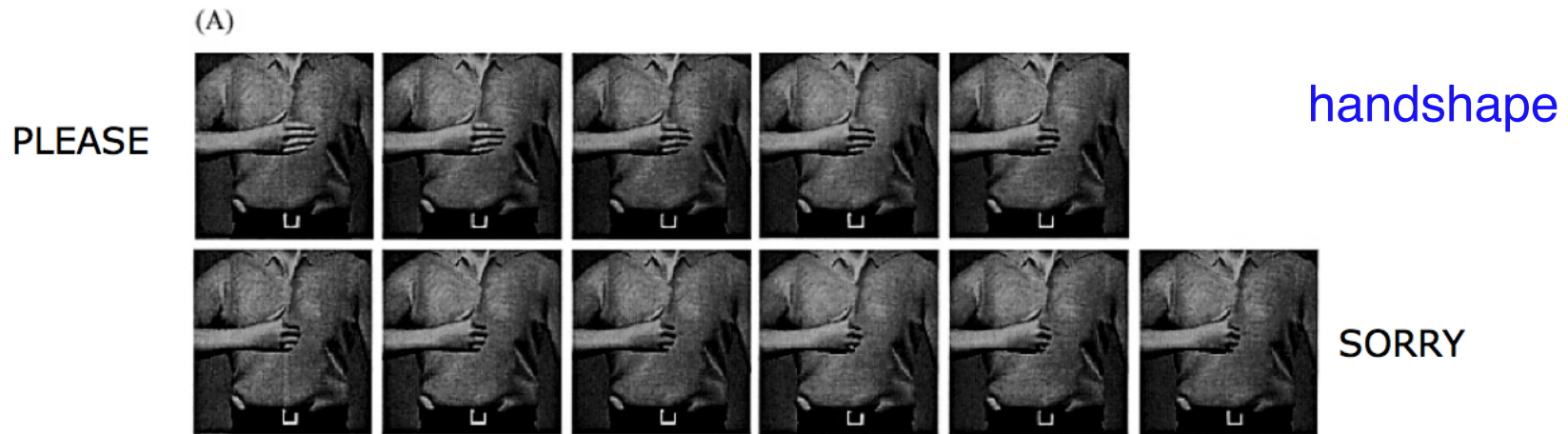
<https://www.handspeak.com/word/search/index.php?id=386>

<https://www.handspeak.com/word/search/index.php?id=393>

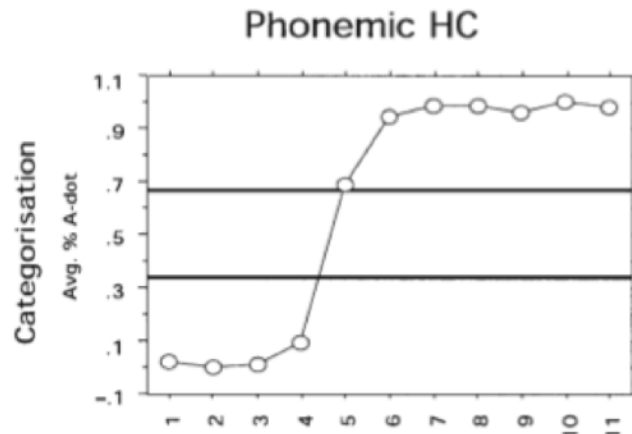


Ex: CHOCOLATE vs. CHURCH

Signers have categorical perception of features



(Emmorey, McCullough, Brentari 2003)



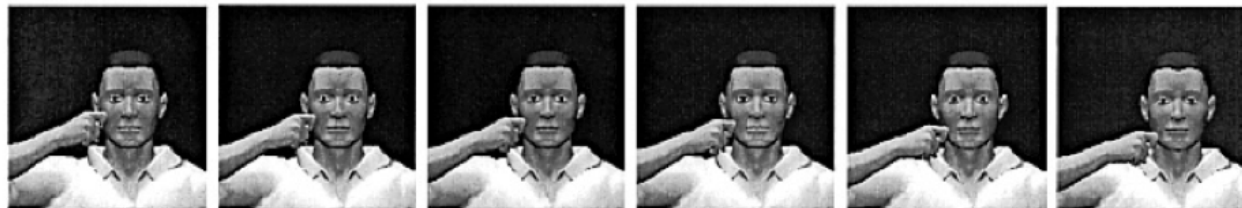
Signers have categorical perception of features

(A)

ONION

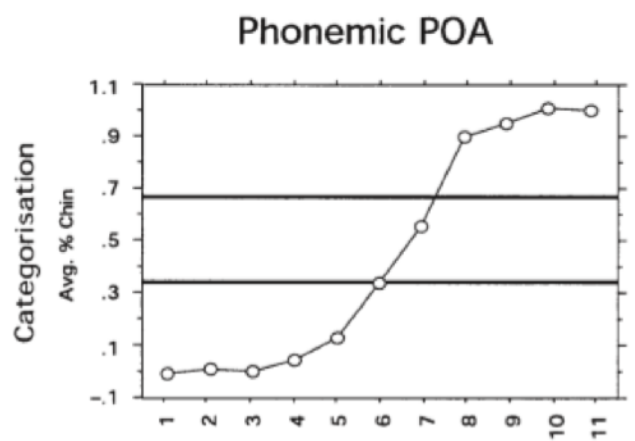


location



APPLE

(Emmorey, McCullough, Brentari 2003)



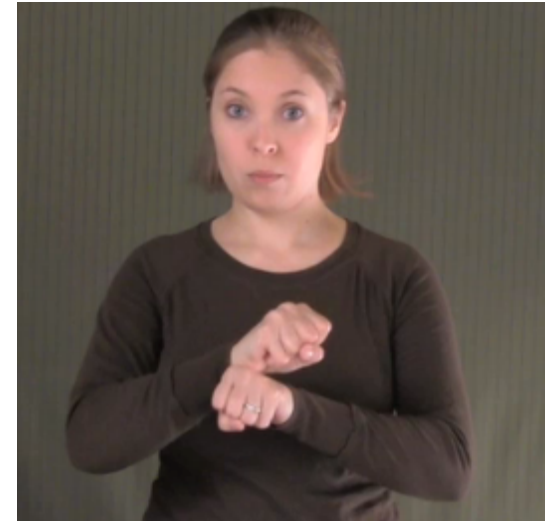
Sign language properties

Sign languages allow for **simultaneous articulation** of information that spoken languages typically don't.

Example: **Aspect** + primary sign

Repeated circular motion: an ongoing, or continuous, event at that point in time [**imperfective**]

Repeated straight motion: a punctuated event that happens multiple times, or habitually [**perfective**]



<http://www.start-american-sign-language.com/sign-language-instruction-asl2-4.html>

Sign language properties

“I'd like to share with you a **piano metaphor**, to have you have a better understanding of how ASL works. So, envision a piano. ASL is broken down into many different grammatical parameters. If you assign a different parameter to each finger as you play the piano -- such as facial expression, body movement, speed, hand shape and so on, as you play the piano -- **English is a linear language, as if one key is being pressed at a time. However, ASL is more like a chord -- all 10 fingers need to come down simultaneously to express a clear concept or idea in ASL.** If just one of those keys were to change the chord, it would create a completely different meaning. The same applies to music in regards to pitch, tone and volume. In ASL, by playing around with these different grammatical parameters, you can express different ideas.” - Christine Sun Kim



[https://www.ted.com/talks/
christine_sun_kim_the_enchanting_music_of_sign_language](https://www.ted.com/talks/christine_sun_kim_the_enchanting_music_of_sign_language)
8:10-9:16

Sign language properties

But there are some similarities in simultaneous articulation...

Example: Y/N question + declarative utterance



Yes/no question face + regular declarative sign sequence

Spoken language equivalent: “echo question”
= Declarative sentence with question prosody

Example: “I can do it?”

The situation

- Deaf individuals aren't all the same
- Deaf parents vs. Hearing parents
 - Deaf-of-deaf children (deaf children of deaf parents) are exposed to a full language immediately
 - Deaf-of-hearing children (deaf children of hearing parents) are exposed to “non-native” signers: they receive inconsistent and incomplete input
- Parents of deaf children also have to make a choice in how to teach their children

Manual / Oral / Total traditions

- Manual tradition
 - Teach sign language exclusively (at least at first)
 - Gives linguistic input from day 1
- Oral tradition
 - Force deaf children to learn spoken language
 - Delayed linguistic input, but potentially better communication with non-signers
- Total communication
 - Expose deaf children both to manual & oral language

Progression of sign language acquisition

Children pass through the same stages as in spoken language acquisition, in the same order: manual babbling to single-sign productions, to multisign combinations, followed by morphological development, more complex syntax, and learning appropriate intonation.



Progression of sign language acquisition

Children make the same kind of mistakes as in spoken language acquisition, such as

- overregularization errors in morphology (“goed”)
- ignoring parental corrections of form
- pronoun reversal errors (confusing what “I” and “you” mean) - despite these being signified by pointing gestures in signed languages, which seems naturally more iconic.



Oral language development

Deaf children are only exposed to lip movements

- This is really hard!

Mouth “Elephant shoes” vs. “I love you.”
vs. “olive juice” vs. “island view”



Oral language development

Phonological development: Deaf children differ during the babble stage from hearing children in both the quality and quantity of sound production. However, some orally trained children develop enough phonological awareness to identify rhymes from lip-reading.

Lexical development: oral vocabulary is delayed and proceeds more slowly.

Syntactic development: delayed, and endpoint of development falls far short of normal language competence.

John goes to fishing.

Him wanted go.

Who TV watched?

Who a boy gave you a ball?

Tom has pushing the wagon.

Oral input only: Impacts on cognitive development



FIGURE 3 | Series of gears in which the first turns clockwise. In which direction will the last gear turn?

Issues with **cognitive offloading**, a key skill that language provides

Deaf children exposed to spoken language only (oralist tradition): “Not only could they not communicate, but they were **left without a code to think in.**”

<https://www.todayifoundout.com/index.php/2010/07/how-deaf-people-think>

Deaf children:

Only an issue when there's impoverished input

Deaf children exposed to sign language learn language the same as normal-hearing children

- There's no inherent deficit in language ability for deaf children

Deaf children exposed to spoken language learn much slower and never catch up to their normal-hearing peers!

- Deficit in spoken language, NOT in language generally



Cochlear Implants

- Cochlear Implants (CI): Allow certain deaf individuals to hear
 - CIs are controversial: treat deafness as a disease which can be “cured”
- How do they work?
 - Replaces the cochlea
 - Takes air pressure and turns it into neural signals



Cochlear implants: Sample speech

8-channel vocoded sentence

Normal sentence



Cochlear implants

- Why are cochlear implants interesting?
 - Explore how oral language develops after a lack of linguistic exposure
 - Effectively allowing these children to be **second language learners of the oral language** (so can potentially use the same approach as we use for investigating the critical/sensitive period with second language learners)



Cochlear implants

- How do these children do with spoken language?
 - **Wide variability**: some catch up to normal-hearing peers, some are unable to use their implants
 - Deficits appear to be due to auditory capabilities (which affects how good the auditory input is)



Cochlear implants

- How do these children do with spoken language?

Note: Musical exposure can help increase spoken language perception for children with cochlear implants:

“Hearing impaired children with cochlear implants who sing regularly have better perception of speech in noise compared to children who don't sing.” - Ritva Torppa, on findings of Torppa, Faulkner, Kujala, Huotilainen, & Lipsanen 2018



<https://www.sciencedaily.com/releases/2018/11/181127111009.htm>

Cochlear implants

An important consideration about cognitive development with cochlear implants:

"The problem is that we can't reliably predict who's going to succeed with the spoken-language approach, and who isn't.

By the time it's clear that a child's spoken language proficiency hasn't supported healthy development across the board, it may be too late for that child to master sign language." — Matthew Hall

<https://www.sciencedaily.com/releases/2016/02/160213185702.htm>



Late exposure to sign language

Delays in syntactic acquisition and language-based analogical reasoning:
Henner, Caldwell-Harris, Novogrodsky, & Hoffmeister 2016

“Studies of Deaf adults have revealed that **late acquisition of sign language is associated with lasting deficits**...Ability in ASL was measured using a syntactic judgment test and language-based analogical reasoning test...Test scores were generally lower for Deaf children who entered the school of assessment after the age of 12...Our results reflect a continuum of outcomes which show that **experience with language is a continuous variable that is sensitive to maturational age.**”



Deaf children: Bigger picture

Implication 1: **Language is a property of the human brain**, not a property predicated on the mouth and ears.

Implication 2: Since deaf children make the same mistakes in learning as hearing children - despite sign languages being more naturally iconic – this suggests that **acquiring a formal grammatical system is a separate cognitive enterprise from learning how to communicate**. If it wasn't, signed languages should be easier to pick up than spoken languages.



Deaf children: Bigger picture

Implication 3: While cochlear implants allow some deaf children to hear spoken language, there is wide variability in the ability to pick up the spoken language. However, **this is a deficit in the spoken modality, rather than a language deficit** - these children still have **native-level proficiency in their signed languages**.



Larger recap: Special Populations

Special populations let us test what matters and what doesn't matter for language acquisition:

- Visual cues: Not crucial for acquiring language (**blind children**)
- Auditory cues: Only crucial for acquiring spoken language (**deaf children**)



Questions?



You should be able to do up through 11 on the review questions and up through 3 on homework 6.